



**Database Systems – Spring 2026**  
**Final Project - Detailed Requirements Document**

**recruitpro**

**A DIGITAL RECRUITMENT MANAGEMENT PLATFORM**

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## 1. Project Overview

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### 1.1 Description

Hiring talent for an organization is a serious problem, the process involve collection of documentations from the applicant through post and manually evaluating bulk of documentation and deciding if a certain applicant is suitable for a position in the organization and if the applicant have enough potential, and if it is actually a good idea to consider him for a position and call him for interview with the competent authorities. It is really a big headache to manually screening candidates evaluating bulk of documentation and it is also difficult for the applicants to share hard copy of all of their documentation to different organizations where he/she wants to apply for job.

RecruitPro is a digital recruitment management platform which solves all the above-mentioned issues to a great extent. The job portals available in the market are just a source of communication between the talent and the organization while our system acts not only a bridge between talent and the organization but it also provides an amazing feature of automatically shortlisting the most relevant candidates for a job role which solve a huge problem of the recruiter and make their job easier.

RecruitPro is a ASP .Net Core MVC Web Application with MySQL server database (which may be hosted as per organization requirements, in this way any one with internet access will have access to data remotely).

### 1.2 Technology Stack

| Layer                | Technology                       |
|----------------------|----------------------------------|
| UI Layer             | HTML, CSS, Javascript            |
| Business Logic Layer | C# controllers                   |
| Data Layer           | C# controllers + MySQL Connector |
| Database             | MySQL                            |
| Version Control      | GitLab                           |

### 1.3 GitLab Repository

<https://gitlab.com/the-real-debuggers/recruit-pro>

## 2. System Architecture

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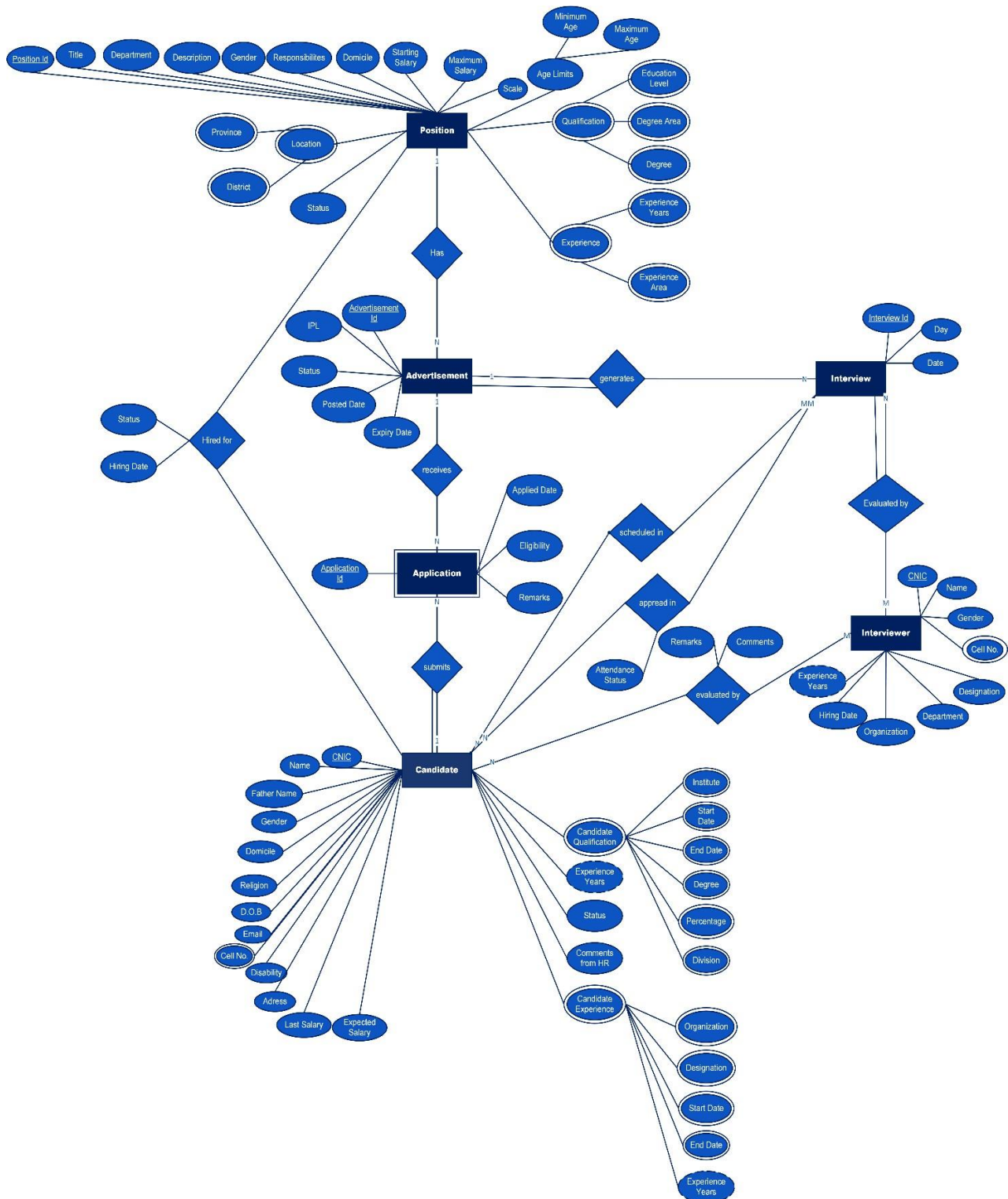
### 2.1 Architecture Overview

The system is an MVC project so its architecture is as follows:

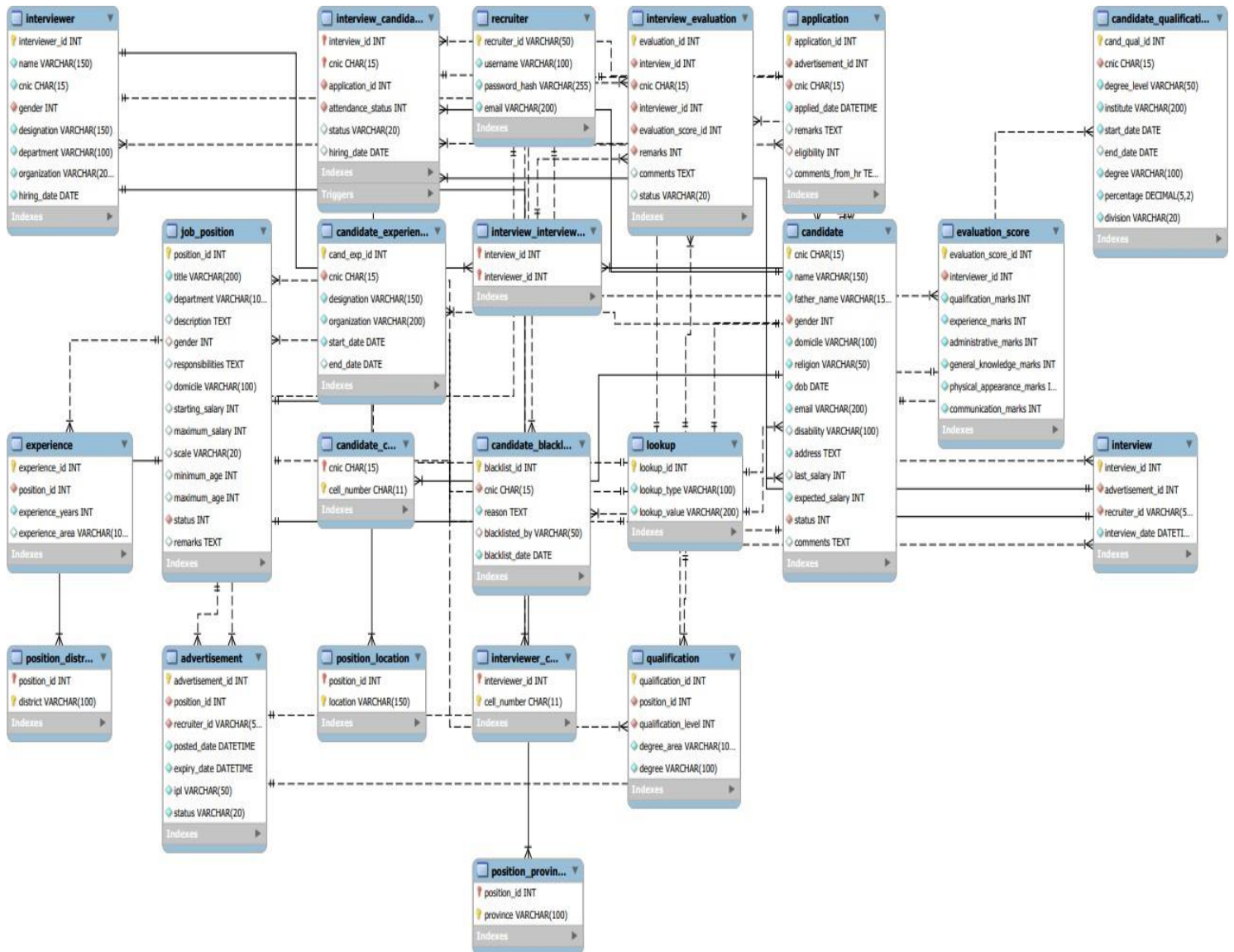
| Tier        | Responsibility                                                                                                                          |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Models      | Responsible for keeping record of core classes of the system which are used for data binding.                                           |
| Views       | Responsible for the frontend of the system                                                                                              |
| Controllers | Responsible for managing user's request, implementation of business logic and coordinating communication between the Model and the View |
| Database    | Tables, constraints, stored procedures, triggers                                                                                        |

### 3. Database Design

#### 3.1 ER Diagram



### 3.2 Schema Summary



### 3.3 Transactions

| #     | Transaction Name       | Tables Affected                                       | Description                                                                                                     |
|-------|------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| TXN-1 | txn_post_advertisement | Job_Position, Advertisement                           | Creates a new Job Position and its Advertisement atomically. Rolls back both if either INSERT fails.            |
| TXN-2 | txn_register_and_apply | Candidate, Application                                | Registers a new Candidate and immediately submits their Application in a single atomic unit.                    |
| TXN-3 | txn_schedule_interview | Interview, Interview_Interviewer, Interview_Candidate | Creates an Interview, assigns an Interviewer, and schedules a Candidate — all three INSERTs in one transaction. |

### 3.4 Views

| #      | View Name                | Source Tables                                                                             | Description                                                                                      |
|--------|--------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| VIEW-1 | vw_open_positions        | Job_Position, Lookup                                                                      | All positions with status = 'Open', including salary range, age limits, and department.          |
| VIEW-2 | vw_active_advertisements | Advertisement, Job_Position, Recruiter                                                    | Active advertisements with position title, recruiter username, and days remaining until expiry.  |
| VIEW-3 | vw_candidate_profile     | Candidate, Lookup                                                                         | Full candidate details with human-readable gender and status labels resolved from Lookup.        |
| VIEW-4 | vw_application_summary   | Application, Candidate, Advertisement, Job_Position, Lookup                               | Each application with candidate name, position title, and eligibility label (or 'Not Assessed'). |
| VIEW-5 | vw_interview_results     | Interview_Candidate, Interview, Candidate, Advertisement, Job_Position, Recruiter, Lookup | Interview outcomes per candidate: attendance, hire status, and hiring date.                      |

### 3.5 Stored Procedures

| #    | Procedure Name              | Parameters                       | Description                                                                                          |
|------|-----------------------------|----------------------------------|------------------------------------------------------------------------------------------------------|
| SP-1 | sp_get_eligible_candidates  | IN: advertisement_id             | Returns all Eligible candidates for a given advertisement, ordered by application date.              |
| SP-2 | sp_blacklist_candidate      | IN: cnic, reason, blacklisted_by | Inserts a Candidate_Blacklist record and updates Candidate.status to 'Blacklisted' in one call.      |
| SP-3 | sp_position_applicant_stats | None                             | Returns per-position statistics: total applications, eligible count, hired count, and pending count. |

### 3.6 Triggers

| #     | Trigger Name          | Event         | Table               | Description                                                                    |
|-------|-----------------------|---------------|---------------------|--------------------------------------------------------------------------------|
| TRG-1 | trg_ic_before_insert  | BEFORE INSERT | Interview_Candidate | Raises an error if status = 'Hired' but hiring_date is NULL. (Original script) |
| TRG-2 | trg_ic_before_update  | BEFORE UPDATE | Interview_Candidate | Same hiring_date validation on UPDATE. (Original script)                       |
| TRG-3 | trg_adv_before_insert | BEFORE INSERT | Advertisement       | Prevents inserting an Advertisement whose expiry_date is in the past.          |
| TRG-4 | trg_app_before_insert | BEFORE INSERT | Application         | Blocks application if Advertisement is not Active OR Candidate is Blacklisted. |



## 4. Features / Modules and Detailed Use Cases:

The system is divided into the following functional modules:

### F1 POSITIONS MANAGEMENT

#### U1.1 Adding a Position

##### Scope: Recruit Pro

##### Level: User Goal

##### Primary Actor: Recruiter

##### Stack Holders and Interests:

Recruiter: Wants to add a new Position.

HR Management: Wants to keep record of Positions in the Organization.

Requesting Department: Wants to confirm the addition of Position and matches with their requirements.

##### Preconditions:

**Recruiter** has been identified and authenticated.

**The position** is approved by the Requesting **Department**, **Human Resource Management** and Competent Authorities.

##### Success Guarantees (Or Post Conditions):

A new Position is created in the system.

##### Main Success Scenario:

1. The **Requesting Department** notifies the **Human Resource Management** about the need for a new Position.
2. The **Human Resource Management** completes the necessary approvals from the competent authorities.
3. The **Human Resource Management** gives a detailed description of Position to the **Recruiter**.
4. The **Recruiter** opens the Positions Management.
5. The **System** shows the list of available Positions in the system.
6. The **Recruiter** creates a new Position.
7. The **System** asks for the details of Position.
8. The **Recruiter** enters the details of Position and save.
9. The **System** creates a new Position.

##### Extensions (or Alternative Flows):

- 9a) The Position of the same detail already exists in the System.
10. The **System** refrains the **Recruiter** from creating the Position.
11. The **Recruiter** changes the details and tries again.

- 9b) The **Recruiter** forgets to add a necessary detail.
10. The **System** will refrain the **Recruiter** from creating the Position
11. The **Recruiter** enters the complete details and tries again.

## U1.2 Editing a Position

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to edit an existing Position.

HR Management: Wants to keep record of updates in Positions in the Organization.

Requesting Department: Wants to confirm that the Position description matches with their requirements.

### Preconditions:

**Recruiter** has been identified and authenticated.

The changes in **Position** are approved by the Requesting **Department**, **Human Resource Management** and Competent Authorities.

### Success Guarantees (Or Post Conditions):

The necessary changes are made in Position's details as per the requirements.

### Main Success Scenario:

1. The **Requesting Department** notifies the **Human Resource Management** about the need for changes in details in an existing Position.
2. The **Human Resource Management** completes the necessary approvals from the competent authorities.
3. The **Human Resource Management** gives a detailed description of Position to the **Recruiter**.
4. The **Recruiter** opens the Positions Module > Positions Management.
5. The **System** shows the list of available Positions in the system.
6. The **Recruiter** selects Edit against the desired Position.
7. The **System** asks for the new details of Position.
8. The **Recruiter** enters the details of Position and save.
9. The **System** creates a new Position.

### Extensions (or Alternative Flows):

8a) The **Recruiter** forgets to add a necessary detail.

9. The **System** will refrain the **Recruiter** from saving the Position.
10. The **Recruiter** enters the complete details and tries again.

## U1.3 InActivating a Position

## Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to inactivate the Position from the system.

HR Management: Wants to keep record of Positions in the Organization.

### Preconditions:

**Recruiter** has been identified and authenticated.

The **Human Resource Management** has approval of removing the Position from the organization.

### Success Guarantees (Or Post Conditions):

The Positions are inactivated.

### Main Success Scenario:

1. The **Requesting Department** notifies the **Human Resource Management** about the need for inactivating an existing Position.
2. The **Human Resource Management** completes the necessary approvals from the competent authorities.
3. The **Human Resource Management** gives a detailed description of Position to the **Recruiter**.
4. The **Recruiter** opens the Positions Management.
5. The **System** shows the list of available Positions in the system.
6. The **Recruiter** selects the desired position and InActivates it.
7. The **System** saves the position as deactivated.

## U1.4 Generate Report for a Position

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to generate a report of the candidates who have applied for a certain position.

HR Management: Wants a specific report related Positions in the Organization.

### Preconditions:

**Recruiter** has been identified and authenticated.

The **position** exists for which the report is to be generated which is needed by the **requested department**.

**Recruiter** has a list of applied candidates for the Position's Advertisement from the **Job Portal**.

### Success Guarantees (Or Post Conditions):

The report containing data of the candidates is generated for a specific post.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** for the details of Applied Candidates of a specific post for a specific department.
2. The **Recruiter** opens Intelligent Screening Module > Import Candidates for the specific position.
3. The **System** shows the list of **Positions** for which candidates have applied.
4. The **Recruiter** searches for the desired **Position**.
5. The **Recruiter** selects the desired **Position**.
6. The **System** shows the Details of the desired **Position**
7. The **Recruiter** selects the spreadsheet to Import.
8. The **System** imports the details of candidates and shows it.

### Extensions (or Alternative Flows):

- 5a) The **Recruiter** does not find the desired **Position**.
6. The **Recruiter** will tell the **Human Resource Management** that the post does not exist anymore.
7. The **Human Resource Management** will look for the actual existing post and inform the **Recruiter**.
8. The **Recruiter** enters the new post and tries again.

## F2 Manage Advertisements

### U2.1 INITIALIZING A NEW Advertisement FOR A POSITION

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to add a new job role in the system.

HR Management: Wants the Advertisement to align with the approved criteria.

#### Preconditions:

**Recruiter** is identified and authenticated.

**Advertisement** has been approved by the Requesting **Department**, **HR Management** and competent authorities.

**The position** is advertised by means of the Communication department.

**Position** Description, Requirements and Criteria has been finalized.

### Success Guarantee (or Post Conditions):

A new **Advertisement** has been initialized against a **Position** in the system.

### Main Success Scenario (or Basic Flow):

1. The **Requesting Department** notifies the **Human Resource Management** that they want to hire resources for a **Position**.
2. The necessary proceedings and approvals are completed.
3. The **Human Resource Management** provides the finalized details, requirements and criteria to Recruiter.
4. The **Recruiter** opens the Positions Module > Advertisements Module.
5. The **System** shows the list of available Positions.
6. The **Recruiter** selects the desired Position.
7. The **System** shows the list of Previous **Advertisements** of that Position.
8. The **Recruiter** “Adds a new Advertisement”.
9. The **System** asks for details of **Advertisement**.
10. The **Recruiter** enters details of **Advertisement** and save.
11. The **System** saves the Advertisement against the **Position**.

### Extensions (or Alternative Flows):

- 6a) The Position does not exist in the system.
7. The **Recruiter** opens the Position Module.
8. The **System** shows the list of Positions.
9. The **Recruiter** “Adds a New Position”.
10. The **System** asks for the details of Position.
11. The **Recruiter** adds the details of Position and save.
12. The **System** saves the Position.
13. The Recruiter continues the remaining steps “INITIALIZING A NEW Advertisement FOR A POSITION”.
- 10a) The Advertisement of the same detail already exists in the System.
11. The **System** refrains the **Recruiter** from initializing the Advertisement
12. The **Recruiter** changes the details and tries again.
- 10b) The **Recruiter** forgets to add a necessary detail.
11. The **System** will refrain the **Recruiter** from initializing the Advertisement.
12. The **Recruiter** enters the complete details and tries again.

### Frequency of Occurrence:

Depending on the requirement of the organization to start a new Advertisement against a Position.

## U2.2 Editing Details of a Advertisement

### Scope: Recruit Pro

### Level: User Goal

## Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to edit a job role in the system.

HR Management: Wants to keep the record of Advertisements up to dated.

### Preconditions:

**Recruiter** is identified and authenticated.

**Advertisement** has been approved by the Requesting **Department**, **HR Management** and competent authorities.

**The position** is advertised by means of the Communication department.

**Position** Description, Requirements and Criteria has been finalized.

### Success Guarantee (or Post Conditions):

A new **Advertisement** has been initialized against a **Position** in the system.

### Main Success Scenario (or Basic Flow):

1. Some changes are required in the Advertisement due to some unavoidable circumstances.
2. The **Human Resource Management** provides the finalized details, requirements and criteria to **Recruiter**.
3. The **Recruiter** opens the **Positions Module > Manage Advertisements**.
4. The **System** shows the list of available Positions.
5. The **Recruiter** selects the desired Position.
6. The **System** shows the list of Previous **Advertisements** of that Position.
7. The **Recruiter** “Edits an Advertisement”.
8. The **System** asks for details of **Advertisement**.
9. The **Recruiter** enters new details of **Advertisement** and saves.
10. The **System** saves the Advertisement against the **Position**.

### Extensions (or Alternative Flows):

- 6a) The Position does not exist in the system.
7. The **Recruiter** opens the Position Module
8. The **System** shows the list of Positions.
9. The **Recruiter** “Adds New Positions”.
10. The **System** asks for the details of Position.
11. The **Recruiter** adds the details of Position and saves.
12. The **System** saves the Position.
13. The Recruiter continues the remaining steps “INITIALIZING A NEW Advertisement FOR A POSITION”.
- 9a) The Advertisement of the same detail already exists in the System.
10. The **System** refrains the **Recruiter** from initializing the Advertisement
11. The **Recruiter** changes the details and tries again.

### Frequency of Occurrence:

Depending on the requirement of the organization to edit details of a Advertisement against a Position.

## U2.3 InActivating an Advertisement

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to inactivate a job role in the system.

HR Management: Wants the Advertisement to be inactivated.

### Preconditions:

**Recruiter** is identified and authenticated.

**Advertisement** has been approved by the Requesting **Department**, **HR Management** and competent authorities and is active.

**The position** is advertised by means of the Communication department.

**Position** Description, Requirements and Criteria has been finalized.

### Success Guarantee (or Post Conditions):

The **Advertisement** has been inactivated against a **Position** in the system.

### Main Success Scenario (or Basic Flow):

1. An Advertisement is required to be inactivated because of some unavoidable circumstances.
2. The **Human Resource Management** provides the finalized details, requirements and criteria to **Recruiter**.
3. The **Recruiter** opens the **Positions Module > Advertisements Module**.
4. The **System** shows the list of available Positions.
5. The **Recruiter** selects the desired Position.
6. The **System** shows the list of Previous **Advertisements** of that Position.
7. The **Recruiter** "InActivates an Advertisement".
8. The **System** InActivates the Advertisement against the **Position**.

### Frequency of Occurrence:

Depending on the requirement of the organization to inactivate an existing Advertisement against a Position.

## U2.4 Generate Report for an Advertisement

### Scope: Recruit Pro

## Level: User Goal

## Primary Actor: Recruiter

## Stack Holders and Interests:

Recruiter: Wants to generate a report of the Advertisement.

HR Management: Wants to a specific report related to advertisements in the organization.

## Preconditions:

**Recruiter** has been identified and authenticated.

The **position** exists for which the report is to be generated which is needed by the **requested department**.

**Recruiter** has a list of applied candidates for the Position's Advertisement from the **Job Portal**.

## Success Guarantees (Or Post Conditions):

The report containing data of the candidates is generated for a specific post.

## Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** for a specific business report regarding Advertisement.
2. The **Recruiter** opens Positions > Manage Advertisement.
3. The **System** shows the list of Advertisements against selected positions.
4. The **Recruiter** searches for the desired **Advertisement**.
5. The **System** shows the Details of the desired **Advertisement**.
6. The **Recruiter** exports the required report and presents it to the Human Resource Management.

## F3 INTELLIGENT SCREENING

### U3.1 IMPORTING CANDIDATES FOR An Advertisement

## Scope: Recruit Pro

## Level: User Goal

## Primary Actor: Recruiter

## Stack Holders and Interests:

Recruiter: Wants to add the data of applied candidates for a Position's Advertisement in the System.

Human Resource Management: Wants details of applied candidates against an Advertisement.

## Preconditions:

**Recruiter** is identified and authenticated.



**Recruiter** has a list of applied candidates for the Position's Advertisement from the **Job Portal**.

The **Position** should exist in the system.

The **Advertisement** should exist in a system.

### Success Guarantee (or Post Conditions):

The details of applied candidates of a specific Advertisement has been imported from a spreadsheet.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** for the details of Applied Candidates of a Advertisement of a Position.
2. The **Recruiter** opens Intelligent Screening Module > Import Candidates for an **Advertisement**.
3. The **System** shows the list of Existing **Advertisements**.
4. The **Recruiter** searches the desired **Advertisement**.
5. The **Recruiter** selects an Advertisement.
6. The **System** shows the Details of Advertisement.
7. The **Recruiter** selects the spreadsheet to Import.
8. The **System** imports the details of candidates and shows it.

### Extensions (or Alternative Flows):

8a) There is some problem in importing candidates.

9. The **Recruiter** notifies the **System Administrator** about the issue.

## U3.2 SHORTLIST CANDIDATES

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to filter out the most relevant candidates.

Human Resource Management: Wants the details of most relevant candidates for a job role.

### Preconditions:

**Recruiter** is identified and authorized.

The **Advertisement** exists in the **System**.

### Success Guarantee (or Post Conditions):

The **System** gives the details of candidates most relevant candidates for a drill.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** for the detail of top most relevant candidates applied for a specific Advertisement a specific Position.
2. The **Recruiter** opens the Intelligent Screening > Screening Candidates.
3. The **System** shows a list of Advertisements available.
4. The **Recruiter** selects an Advertisement.
5. The **System** displays the list of most relevant candidates and allows the Recruiter to apply further filtering.
6. The **Recruiter** creates a list of Final shortlisted candidates for the **Advertisement** and save.
7. The **System** maintains the record of eligible candidates.
8. The **Recruiter** communicates the details with **Human Resource Management**.

### U3.2.1 Generate Screening Summary for an Advertisement

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to generate a screening summary of the Advertisement.

HR Management: Wants a specific summary related to Advertisement.

Requesting Department: Wants to keep all valid data of Advertisements for a specific post so they may continue to shortlist the candidates based on merit.

#### Preconditions:

**Recruiter** has been identified and authenticated.

The **Advertisement** exists for which the summary is to be generated which is needed by the **requested department**.

**Recruiter** has a list of applied candidates for the Position's Advertisement from the **Job Portal**.

#### Success Guarantees (Or Post Conditions):

The report containing data of the candidates is generated for a specific post.

#### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** for generating a screening summary for an Advertisement of a specific post for a specific department.
2. The **Recruiter** opens Intelligent Screening Module > view an Advertisement for the specific position.
3. The **Recruiter** searches for the desired **Advertisement**.
4. The **Recruiter** selects the desired **Advertisement**.
5. The **System** shows the Details of the desired **Advertisement**.
6. The **Recruiter** generates the screening summary of the required Advertisement.
7. The **System** imports the summary and shows it.

#### Extensions (or Alternative Flows):

- 5a) The **Recruiter** does not find the desired **Advertisement**.
- 6. The **Recruiter** will tell the **Human Resource Management** that the Advertisement does not exist.
- 7. The **Human Resource Management** will look for the actual existing Advertisement and inform the **Recruiter**.
- 8. The **Recruiter** enters the actual Advertisement and tries again.

## U3.3 Calculate Experience

### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to calculate the experience of a candidate manually before selecting him for the interview.

#### Preconditions:

**Recruiter** is identified and authorized.

The candidate is physically present with their experience letters.

#### Main Success Scenario (or Basic Flow):

- 1. The candidate appears in an Interview.
- 2. The **Human Resource Management** wants to manually verify the experiences of candidates from their experience letters before selecting him for the interview.
- 3. The **Recruiter** asks the candidates for his experience letters.
- 4. The **Recruiter** opens the Intelligent Screening Engine > Experience Calculator.
- 5. The **Recruiter** enters the details of the candidate's experiences.
- 6. The **System** calculates the total experience of the candidate and shows it.
- 7. The **Recruiter** lets the candidate appear in the interview.

## F4 Interviews Management

### U4.1 Schedule Interviews

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to assign different days to eligible candidates, and assign the interview committee.  
Human Resource Management: Wants to interview different candidates on different days.

### Preconditions:

**Recruiter** is identified and authorized.

The Candidates are finalized for the Advertisement.

The Interview days and Interview Committee is finalized by the **Human Resource Management**.

### Main Success Scenario (or Basic Flow):

1. The Human Resource Management communicates the Advertisement Interviews Days and Committee with the Recruiter.
2. The **Recruiter** opens Interview Management > Schedule Interview
3. The **Systems** shows a list of Currently Active Advertisements.
4. The **Recruiter** selects an Advertisement.
5. The System asks the Recruiter to add the desired number of days.
6. The **Recruiter** adds a day, selects dates and interview committee for that day.
7. The **Recruiter** repeats Step 6 for each day to be added.

### Extensions (or Alternative Flows):

6a) A member of the Interview Committee is not available in the system.

7. The **Recruiter** Adds the Interviewer in the system and then continues

## U4.2 Manage Interviewers

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to keep record of Interview Panel Members (Interviewers).

### Preconditions:

**Recruiter** is identified and authorized.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** communicates to the Recruiter with the details of the Interviewer.
2. The **Recruiter** opens Interviews Management > Manage Interviewers.
3. The **System** shows the list of existing Interviewers.
4. The **Recruiter** Adds a new Interview / Delete an existing Interviewer or edit an existing Interviewer details.
5. The **System** saves the changes.

### Extensions (or Alternative Flows):

- 4a) The **Recruiter** forgets to add a necessary detail of an Interviewer.
5. The **System** refrains the Recruiter from saving the changes or adding a new Interviewer.
6. The **Recruiter** adds the remaining details and continues.

## U4.3 Assign Interview Days to Candidates

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to assign different days to candidates selected for a certain Advertisement.

Human Resource Management: Wants to interview different candidates on different days.

Interviewers: Wants details of the candidates to interview.

### Preconditions:

**Recruiter** is identified and authorized.

The Interviews are scheduled.

The Interview Committee and Interview dates are finalized.

The candidates have been finalized for the Advertisement.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the Recruiter to divide the finalized candidates for an Advertisement into different days.
2. The **Recruiter** opens the Interviews Management > Assign Interview Days.
3. The **System** shows a list of currently existing Advertisements.
4. The **Recruiter** selects the desired Advertisement.
5. The **System** shows the list of finalized candidates for an Advertisement.
6. The **Recruiter** assigns different Interview days to candidates.

## F5 Candidates Management

### U5.1.1 Manage Candidates

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to view/edit details of a candidate.

Human Resource Management: Wants to keep an accurate record of candidates.

### **Preconditions:**

**Recruiter** is identified and authorized.

### **Main Success Scenario (or Basic Flow):**

1. The **Human Resource Management** asks the **Recruiter** for details of a specific candidate.
2. The **Recruiter** opens the Candidate Management.
3. The **System** shows a list of all existing candidates
4. The **Recruiter** searches the desired candidates and views, edits its profile.
5. The **Recruiter** can also add a candidate when needed.

#### **U5.1.2 Edit Candidates**

### **Scope: Recruit Pro**

### **Level: User Goal**

### **Primary Actor: Recruiter**

### **Stack Holders and Interests:**

Recruiter: Wants to edit details of a candidate.

Human Resource Management: Wants to keep an accurate record of candidates.

### **Preconditions:**

**Recruiter** is identified and authorized.

### **Main Success Scenario (or Basic Flow):**

1. The **Human Resource Management** asks the **Recruiter** for details of a specific candidate.
2. The **Recruiter** opens the Candidate Management.
3. The **System** shows a list of all existing candidates
4. The **Recruiter** searches the desired candidates and views, edits its profile.

#### **U5.1.3 Generate Report of a Candidate**

### **Scope: Recruit Pro**

### **Level: User Goal**

### **Primary Actor: Recruiter**

### **Stack Holders and Interests:**

Recruiter: Wants to generate a report of a candidate.

HR Management: Wants details of a candidate

### Preconditions:

**Recruiter** has been identified and authenticated.

**Recruiter** has a list of applied candidates for the Position's Advertisement from the **Job Portal**.

### Success Guarantees (Or Post Conditions):

The report containing data of the candidate is generated for a specific post.

### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** to generate reports of Candidates who have applied for a specific post for a specific department.
2. The **Recruiter** searches for the desired candidate.
3. The **System** shows the Details of the desired **Candidate**.
4. The **Recruiter** imports spreadsheet of the Candidate.
5. The **System** shows the Details of the desired **Candidate**.

### Extensions (or Alternative Flows):

- 5a) The **Recruiter** does not find the desired **Candidate**.
6. The **Recruiter** will tell the **Human Resource Management** that the **Candidate** does not exist.
7. The **Human Resource Management** will look for the **Candidate** and inform the **Recruiter**.
8. The **Recruiter** enters the new **Candidate** and tries again.

## U5.2 Blacklist Candidate

### Scope: Recruit Pro

### Level: User Goal

### Primary Actor: Recruiter

### Stack Holders and Interests:

Recruiter: Wants to keep record of blacklisted candidates so that he does not shortlist the candidate for any Advertisement.

Human Resource Management: Wants to blacklist a candidate because of some specific reasons.

### Preconditions:

**Recruiter** is identified and authorized.

The Human Resource Management has notified the **Recruiter** to blacklist a candidate.

## Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the Recruiter to blacklist a candidate for some reasons and never shortlist him for upcoming Advertisements of any position.
2. The **Recruiter** opens Candidate Management
3. The **System** shows a list of all candidates.
4. The **Recruiter** adds another candidate into the blacklist.
5. The **System** shows the list of all existing candidates.
6. The **Recruiter** searches the candidate and adds him/her to the blacklist.
7. The **System** adds the candidate to the blacklist.

### U5.2.1 Generate Report of Blacklisted Candidates

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to generate a report of the blacklisted candidates.

HR Management: Wants to keep the record of blacklisted candidates in the Organization.

Requesting Department: Wants to keep all valid data of blacklisted applicants for a specific post so they may continue to shortlist the candidates based on merit.

#### Preconditions:

**Recruiter** has been identified and authenticated.

The **blacklisted candidate** exists for which the report is to be generated which is needed by the **requested department**.

**Recruiter** has a list of **blacklisted** candidates for the Position's Advertisement from the **Job Portal**.

#### Success Guarantees (Or Post Conditions):

The report containing data of the **blacklisted** candidate is generated for a specific post.

## Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the **Recruiter** to generate reports of blacklisted Candidates who have applied for a specific post for a specific department.
2. The **Recruiter** opens Intelligent Screening Module > Import blacklisted Candidates for the specific position.
3. The **System** shows the list of **Positions** for which candidates have applied.
4. The **Recruiter** searches for the desired **Position**.
5. The **Recruiter** selects the desired **Position**.
6. The **System** shows the Details of the desired **Position**.
7. The **Recruiter** searches for the desired blacklisted candidates.
8. The **System** shows the Details of the desired **blacklisted Candidates**.
9. The **Recruiter** imports spreadsheet of the blacklisted Candidates.



10. The **System** shows the Details of the desired blacklisted **Candidate**.

### Extensions (or Alternative Flows):

5a) The **Recruiter** does not find the desired blacklisted **Candidates**.

6. The **Recruiter** will tell the **Human Resource Management** that the blacklisted **Candidates** do not exist.

7. The **Human Resource Management** will look for the blacklisted **Candidates** and inform the **Recruiter**.

8. The **Recruiter** enters the new blacklisted **Candidate** and tries again.

## F6 Candidates Attendance

### U6.1 Mark Attendance

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Recruiter

#### Stack Holders and Interests:

Recruiter: Wants to mark attendance of the candidates who appeared on a specific day of a specific Advertisement.

Human Resource Management: Wants to keep record of candidates appearing on a specific day of a specific Advertisement.

#### Preconditions:

**Recruiter** is identified and authorized.

Interviews have been scheduled for a specific Advertisement.

#### Main Success Scenario (or Basic Flow):

1. The **Human Resource Management** asks the Recruiter to mark the attendance of candidates appearing on a specific day of a specific Advertisement.
2. The **Recruiter** opens Candidates Attendance > Mark Attendance.
3. The **System** asks the user to Select a Position and an Advertisement.
4. The **Recruiter** selects a Position and an Advertisement.
5. The **System** shows the list of Days of that Advertisement.
6. The **Recruiter** selects a Day.
7. The **System** shows the list of candidates whose interview has been scheduled on that day of an Advertisement.
8. The **Recruiter** marks the Attendance of candidates and saves.
9. The **System** saves the Attendance of candidates against the specific day of specific Advertisement.

## F7 Candidates Evaluation

### U7.1 Interview Candidate

#### Scope: Recruit Pro

#### Level: User Goal

#### Primary Actor: Interviewer

#### Stack Holders and Interests:

Interviewer: Wants to mark the evaluation of a Candidate.

Human Resource Management Wants to keep record of remarks of interview committee for a specific candidate against a specific Advertisement.

#### Preconditions:

The interviewer is identified and authenticated.

The interviewer is a panel member on a specific day of a specific Advertisement.

#### Main Success Scenario (or Basic Flow):

1. The **Interviewer** opens the Candidate Evaluation > Interview Candidate.
2. The **System** shows a list of Advertisements and days in which the **Interviewer** is a panel member.
3. The **Interviewer** selects the desired day of an Advertisement.
4. The **System** shows the list of candidates accordingly.
5. The **Recruiter** views the Profile of the candidate sitting in front, interviews the candidate and marks the evaluation accordingly and save.
6. The system saves the Interviewer Remarks.

## 5. Use Cases

| Features or Modules          | Use case ID | Use Case Name                                   | Interface Id   |
|------------------------------|-------------|-------------------------------------------------|----------------|
| <b>Positions Management</b>  | U1.1        | Adding a Position                               | I2.1.1         |
|                              | U1.2        | Editing a Position                              |                |
|                              | U1.3        | Inactivating a Position                         |                |
|                              | U1.4        | Generate Report for a Position                  |                |
| <b>Manage Advertisements</b> | U2.1        | Initializing a new Advertisement for a Position | I3.1.1         |
|                              | U2.2        | Editing Details of an Advertisement             |                |
|                              | U2.3        | InActivating an Advertisement                   |                |
|                              | U2.4        | Generate Report of an Advertisement             |                |
| <b>Intelligent Screening</b> | U3.1        | Import Candidates for an Advertisement          | I4.1.1         |
|                              | U3.2        | Shortlist Candidates                            | I4.2.1         |
|                              | U3.2.1      | Generate Screening Summary for an Advertisement |                |
|                              | U3.3        | Calculate Experience                            |                |
| <b>Interviews Management</b> | U4.1        | Schedule Interview                              | I5.1.1, I5.1.2 |
|                              | U4.2        | Manage Interviewers                             |                |
|                              | U4.3        | Assign Different Days to Candidates             | I5.1.3         |
| <b>Candidates Management</b> | U5.1.1      | Manage Candidates                               | I6.1.1         |
|                              | U5.1.2      | Edit Candidates                                 |                |
|                              | U5.1.3      | Generate Report of a Candidate                  |                |
|                              | U5.2        | Blacklist Candidates                            |                |
|                              | U5.2.1      | Generate Report of Blacklisted Candidates       |                |
| <b>Candidates Attendance</b> | U6.1        | Mark Attendance                                 | I7.1.1, I7.1.2 |

|                          |      |                     |                |
|--------------------------|------|---------------------|----------------|
| Candidates<br>Evaluation | U7.1 | Interview Candidate | I8.1.1, I8.1.2 |
|--------------------------|------|---------------------|----------------|

## 6. Interface Design

Each interface section includes a screenshot and a list of validators.

### I1.1.1 Dashboard

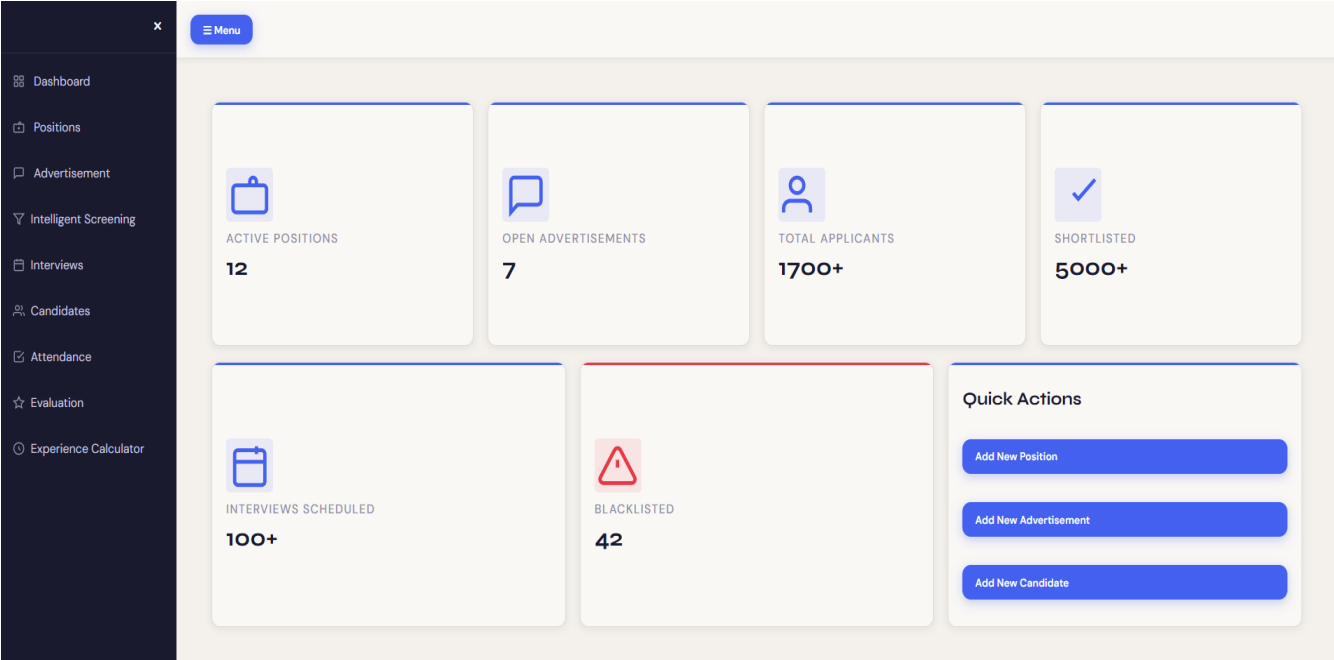
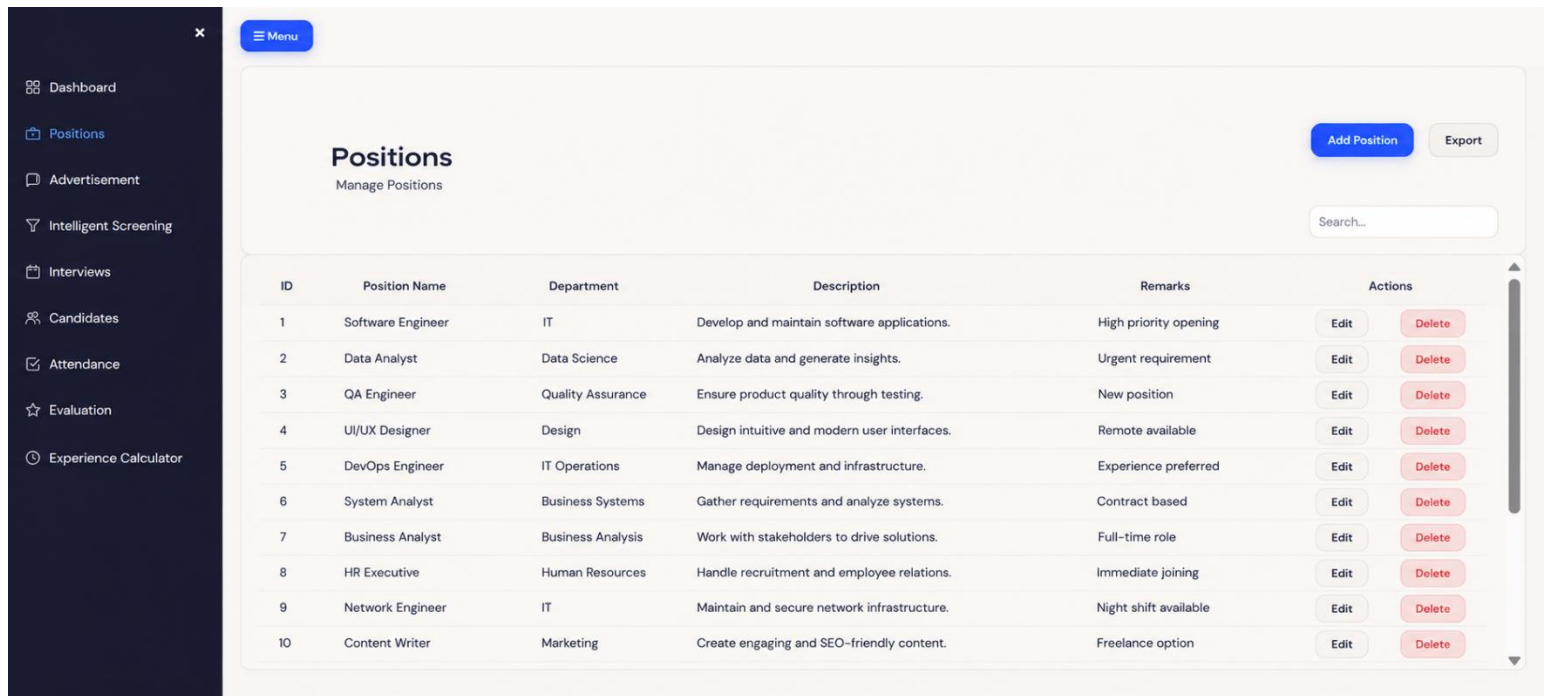


Figure 1 shows Dashboard

## I2.1.1 Managing Positions



**Positions**  
Manage Positions

[Add Position](#) [Export](#)

Search...

| ID | Position Name     | Department        | Description                                  | Remarks               | Actions                                     |
|----|-------------------|-------------------|----------------------------------------------|-----------------------|---------------------------------------------|
| 1  | Software Engineer | IT                | Develop and maintain software applications.  | High priority opening | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2  | Data Analyst      | Data Science      | Analyze data and generate insights.          | Urgent requirement    | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3  | QA Engineer       | Quality Assurance | Ensure product quality through testing.      | New position          | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4  | UI/UX Designer    | Design            | Design intuitive and modern user interfaces. | Remote available      | <a href="#">Edit</a> <a href="#">Delete</a> |
| 5  | DevOps Engineer   | IT Operations     | Manage deployment and infrastructure.        | Experience preferred  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 6  | System Analyst    | Business Systems  | Gather requirements and analyze systems.     | Contract based        | <a href="#">Edit</a> <a href="#">Delete</a> |
| 7  | Business Analyst  | Business Analysis | Work with stakeholders to drive solutions.   | Full-time role        | <a href="#">Edit</a> <a href="#">Delete</a> |
| 8  | HR Executive      | Human Resources   | Handle recruitment and employee relations.   | Immediate joining     | <a href="#">Edit</a> <a href="#">Delete</a> |
| 9  | Network Engineer  | IT                | Maintain and secure network infrastructure.  | Night shift available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 10 | Content Writer    | Marketing         | Create engaging and SEO-friendly content.    | Freelance option      | <a href="#">Edit</a> <a href="#">Delete</a> |

Figure 2 shows Positions Management module

### Validators List:

- Position Name: Must not be empty; no special characters allowed
- Position Code: Unique identifier; alphanumeric only
- Search Field: Accepts text input; trims whitespace
- Insert New Position Button: Disabled if required fields are incomplete

### 13.1.1 Initializing a new Advertisement for a Position

| ID | Advertisement Title                 | Position                       | Description                                                                          | Status  | Posted On    | Actions                                     |
|----|-------------------------------------|--------------------------------|--------------------------------------------------------------------------------------|---------|--------------|---------------------------------------------|
| 1  | Experienced Java Developer Needed   | Java Developer                 | We are looking for an experienced Java Developer to build scalable web applications. | Active  | May 15, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2  | Marketing Executive – Join Our Team | Marketing Executive            | Seeking a creative and result-driven Marketing Executive for brand promotion.        | Active  | May 12, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3  | Data Analyst Vacancy                | Data Analyst                   | Analyze data and generate insights to support business decisions.                    | Closed  | May 10, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4  | UI/UX Designer Required             | UI/UX Designer                 | We need a passionate UI/UX Designer to create intuitive user experiences.            | Active  | May 8, 2024  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 5  | HR Manager – Lead Our People        | HR Manager                     | Looking for an experienced HR Manager to lead our HR department.                     | Active  | May 5, 2024  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 6  | Business Development Executive      | Business Development Executive | Identify new business opportunities and build strong client relationships.           | Expired | Apr 30, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 7  | Customer Support Specialist         | Customer Support Specialist    | Provide excellent support and resolve customer queries in a timely manner.           | Closed  | Apr 28, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 8  | DevOps Engineer Opening             | DevOps Engineer                | Looking for a DevOps Engineer to manage CI/CD pipelines and cloud infrastructure.    | Active  | Apr 25, 2024 | <a href="#">Edit</a> <a href="#">Delete</a> |

Figure 3 shows Advertisement Management module

#### Validators List:

- Advertisement Title: Must not be empty
- Select Position Dropdown: A valid position must be selected before inserting
- Search Field: Accepts text input; trims whitespace
- Insert New Advertisement Button: Disabled if no position is selected

### 14.1.1 Import Candidates for an Advertisement

**Intelligent Screening**

Available Advertisements

Import Candidate

**Details of Imported Candidates**

| ID | Candidate Name | Email                  | Phone        | Position Applied            | Source File              | Imported On           | Status   | Actions                                     |
|----|----------------|------------------------|--------------|-----------------------------|--------------------------|-----------------------|----------|---------------------------------------------|
| 1  | Ayesha Khan    | ayeshakhan@email.com   | 0300-1234567 | Software Engineer           | Candidates_May15.xlsx    | May 15, 2024 10:30 AM | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 2  | Bilal Ahmed    | bilalahmed@email.com   | 0311-2345678 | Data Analyst                | Data_Analysts_May12.xlsx | May 12, 2024 02:15 PM | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 3  | Fatima Noor    | fatima.noor@email.com  | 0321-3456789 | Marketing Executive         | Marketing_Team.xlsx      | May 10, 2024 11:45 AM | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 4  | Usman Ali      | usman.ali@email.com    | 0333-4567890 | UI/UX Designer              | Designers_April.xlsx     | May 8, 2024 09:20 AM  | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 5  | Hina Butt      | hina.butt@email.com    | 0345-5678901 | HR Manager                  | HR_Candidates.xlsx       | May 5, 2024 03:10 PM  | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 6  | Omar Farooq    | omar.farooq@email.com  | 0302-6789012 | DevOps Engineer             | Tech_Team_April.xlsx     | Apr 30, 2024 10:05 AM | Warning  | <a href="#">View</a> <a href="#">Remove</a> |
| 7  | Sana Raza      | sana.raza@email.com    | 0314-7890123 | Business Analyst            | Business_Analysts.xlsx   | Apr 28, 2024 04:25 PM | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 8  | Zain Abbas     | zain.abbas@email.com   | 0322-8901234 | Customer Support Specialist | Support_Team.xlsx        | Apr 25, 2024 01:30 PM | Failed   | <a href="#">View</a> <a href="#">Remove</a> |
| 9  | Mariam Iqbal   | mariam.iqbal@email.com | 0336-9012345 | QA Engineer                 | QA_Team.xlsx             | Apr 22, 2024 09:50 AM | Imported | <a href="#">View</a> <a href="#">Remove</a> |
| 10 | Danish Malik   | danish.malik@email.com | 0308-0123456 | Content Writer              | Content_Writers.xlsx     | Apr 20, 2024 11:10 AM | Warning  | <a href="#">View</a> <a href="#">Remove</a> |

Figure 4 shows Intelligent Screening module

#### Validators List:

- Advertisement Selection: A valid advertisement must exist in the list
- Search Field: Accepts text input; trims whitespace
- Import Candidates Button: Only active when advertisement is selected
- Shortlist Candidates Button: Only active when advertisement is selected
- Spreadsheet File: Must be a valid Excel (.xlsx/.xls) file
- Select Spread Sheet to Import Dropdown: A file must be selected before importing
- Imported Data: Validated for required columns (Name, CNIC, etc.); duplicates flagged
- Back Button: Navigates to previous screen without saving



### 14.1.2 Shortlist Candidates for an Advertisement

**Intelligent Screening** Export Options

**Shortlist Candidates**

**All Candidates** | Eligible Candidates | Not Eligible Candidates

Qualification ▼ Age ▼ Gender ▼ Domicile ▼ Experiences ▼ Experience Years ▼ Search...

| ID | Candidate Name | Qualification             | Age | Gender | Domicile   | Experience          | Experience Years | Profile              | Eligibility    | Remarks                 |
|----|----------------|---------------------------|-----|--------|------------|---------------------|------------------|----------------------|----------------|-------------------------|
| 1  | Ayesha Khan    | BS Computer Science       | 26  | Female | Lahore     | Software Engineer   | 2.5 Years        | <a href="#">View</a> | Eligible ▼     | Strong technical skills |
| 2  | Bilal Ahmed    | BS Data Science           | 28  | Male   | Karachi    | Data Analyst        | 3 Years          | <a href="#">View</a> | Eligible ▼     | Good analytical skills  |
| 3  | Fatima Noor    | MBA Marketing             | 27  | Female | Islamabad  | Marketing Executive | 2 Years          | <a href="#">View</a> | Not Eligible ▼ | Less experience         |
| 4  | Usman Ali      | BS Software Engineering   | 25  | Male   | Faisalabad | UI/UX Designer      | 1.8 Years        | <a href="#">View</a> | Eligible ▼     | Creative portfolio      |
| 5  | Hina Butt      | BBA (Hons)                | 24  | Female | Rawalpindi | HR Executive        | 1.5 Years        | <a href="#">View</a> | Eligible ▼     | Good communication      |
| 6  | Omar Farooq    | BS Information Technology | 30  | Male   | Peshawar   | DevOps Engineer     | 4 Years          | <a href="#">View</a> | Not Eligible ▼ | Missing certification   |
| 7  | Sana Raza      | BS Business Admin         | 26  | Female | Multan     | Business Analyst    | 2.2 Years        | <a href="#">View</a> | Eligible ▼     | Relevant experience     |
| 8  | Zain Abbas     | BS Computer Science       | 27  | Male   | Lahore     | Software Engineer   | 3.3 Years        | <a href="#">View</a> | Eligible ▼     | Good problem solving    |
| 9  | Mariam Iqbal   | MA English                | 25  | Female | Sialkot    | Content Writer      | 1.2 Years        | <a href="#">View</a> | Not Eligible ▼ | Does not meet criteria  |
| 10 | Danish Malik   | BS Cyber Security         | 29  | Male   | Karachi    | Security Analyst    | 3 Years          | <a href="#">View</a> | Eligible ▼     | Strong security skills  |

Figure 5 shows Import Candidates for an Advertisement submodule

#### Validators List:

- Qualification Filter: Must match predefined qualification categories
- Age Filter: Must be a valid numeric range
- Gender Filter: Must select from allowed options (Male/Female/All)
- Domicile Filter: Must select from predefined domicile list
- Experiences Filter: Must select from predefined experience categories
- Experience Years Filter: Must be a valid numeric range
- Eligibility Dropdown per Candidate: Only “Eligible” or “Not Eligible” allowed
- Remarks Field: Optional; text input
- Search Field: Accepts text input; trims whitespace

### I5.1.1 Schedule Interview

**Interviews**  
Manage Interviews

Schedule Interview Export

Search...

| ID  | Candidate Name | Position               | Interview Date | Time     | Interviewer        | Type      | Status      | Actions                |
|-----|----------------|------------------------|----------------|----------|--------------------|-----------|-------------|------------------------|
| 101 | Ayesha Khan    | Software Engineer      | 15 May 2025    | 10:00 AM | Mr. Ahmed Raza     | Technical | Scheduled   | View Reschedule Cancel |
| 102 | Usman Ali      | System Analyst         | 16 May 2025    | 11:30 AM | Ms. Sana Malik     | HR        | Confirmed   | View Reschedule Cancel |
| 103 | Hina Fatima    | DevOps Engineer        | 17 May 2025    | 02:00 PM | Mr. Bilal Ahmed    | Technical | Scheduled   | View Reschedule Cancel |
| 104 | Mohammad Hamza | UI/UX Designer         | 18 May 2025    | 09:00 AM | Ms. Iqra Naveed    | Technical | Confirmed   | View Reschedule Cancel |
| 105 | Zara Noor      | QA Engineer            | 19 May 2025    | 01:00 PM | Mr. Faisal Khan    | Technical | Rescheduled | View Reschedule Cancel |
| 106 | Ali Raza       | Database Administrator | 20 May 2025    | 03:30 PM | Ms. Mahnoor Ali    | Technical | Completed   | View Reschedule Cancel |
| 107 | Sana Javed     | Product Manager        | 21 May 2025    | 10:30 AM | Mr. Omer Farooq    | HR        | Scheduled   | View Reschedule Cancel |
| 108 | Danish Ahmed   | Business Analyst       | 22 May 2025    | 12:00 PM | Ms. Rabia Saleem   | HR        | Confirmed   | View Reschedule Cancel |
| 109 | Maryam Hashmi  | Mobile App Developer   | 23 May 2025    | 04:00 PM | Mr. Noman Siddiqui | Technical | Cancelled   | View Reschedule Cancel |
| 110 | Farhan Saeed   | Network Engineer       | 24 May 2025    | 11:00 AM | Ms. Hira Ahmed     | Technical | Scheduled   | View Reschedule Cancel |

Figure 6 shows Interviews Management module

#### Validators List:

- Advertisement Selection: A valid advertisement must be listed
- Search Field: Accepts text input; trims whitespace
- Schedule Interviews Button: Only active when an advertisement row is selected

## 15.1.2 Schedule Interview

The screenshot displays the 'Interviews Management' interface. On the left is a dark sidebar with navigation links: Dashboard, Positions, Advertisement, Intelligent Screening, Interviews (highlighted), Candidates, Attendance, Evaluation, and Experience Calculator. The main content area has a 'Schedule Interview' subheader, a '+ Add Interview Day' button, a search bar, and an 'Export' button. Below this is a table with 7 rows of interview data.

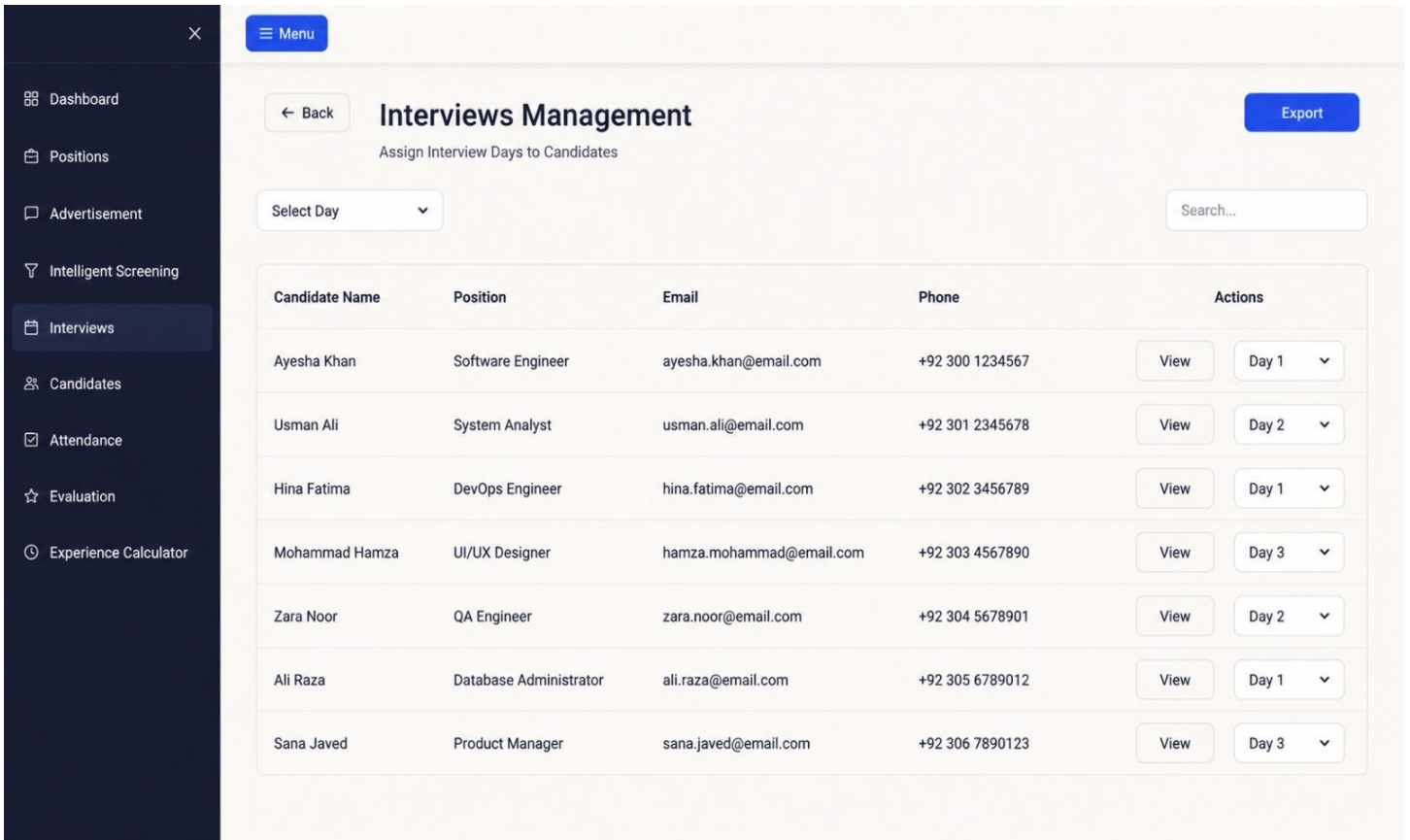
| ID | Interview Day               | Date        | Start Time | End Time | Location          | Total Interviews | Actions |      |                   |
|----|-----------------------------|-------------|------------|----------|-------------------|------------------|---------|------|-------------------|
| 1  | Day 1 - Technical Screening | 15 May 2025 | 09:00 AM   | 01:00 PM | Conference Room A | 12               | View    | Edit | Assign Candidates |
| 2  | Day 2 - HR Round            | 16 May 2025 | 10:00 AM   | 02:00 PM | Meeting Room 2B   | 10               | View    | Edit | Assign Candidates |
| 3  | Day 3 - Technical Interview | 17 May 2025 | 09:30 AM   | 03:30 PM | Conference Room A | 14               | View    | Edit | Assign Candidates |
| 4  | Day 4 - Final Interview     | 19 May 2025 | 11:00 AM   | 04:00 PM | Board Room        | 8                | View    | Edit | Assign Candidates |
| 5  | Day 5 - Special Round       | 20 May 2025 | 02:00 PM   | 06:00 PM | Conference Room B | 6                | View    | Edit | Assign Candidates |
| 6  | Day 6 - Backup Round        | 22 May 2025 | 09:00 AM   | 01:00 PM | Meeting Room 3C   | 7                | View    | Edit | Assign Candidates |
| 7  | Day 7 - Final Backup        | 23 May 2025 | 10:00 AM   | 02:00 PM | Conference Room A | 5                | View    | Edit | Assign Candidates |

Figure 7 shows Schedule Interview submodule

### Validators List:

- Add Interview Day Button: At least one day required before assigning candidates
- Interview Day Date: Must be a valid calendar date; cannot be in the past
- Edit Button: Only enabled when an interview day row is selected
- Assign Candidates Button: Only enabled when a valid interview day exists
- Search Field: Accepts text input; trims whitespace

### 15.1.3 Assign Different Days to Candidates



The screenshot displays the 'Interviews Management' submodule. It features a sidebar with navigation options: Dashboard, Positions, Advertisement, Intelligent Screening, Interviews (selected), Candidates, Attendance, Evaluation, and Experience Calculator. The main content area has a 'Menu' button and a 'Back' button. The title 'Interviews Management' is followed by the subtitle 'Assign Interview Days to Candidates'. There is an 'Export' button and a 'Select Day' dropdown menu. A search bar is also present. The main table lists candidates with their details and assigned interview days.

| Candidate Name | Position               | Email                    | Phone           | Actions |         |
|----------------|------------------------|--------------------------|-----------------|---------|---------|
| Ayesha Khan    | Software Engineer      | ayesha.khan@email.com    | +92 300 1234567 | View    | Day 1 ▼ |
| Usman Ali      | System Analyst         | usman.ali@email.com      | +92 301 2345678 | View    | Day 2 ▼ |
| Hina Fatima    | DevOps Engineer        | hina.fatima@email.com    | +92 302 3456789 | View    | Day 1 ▼ |
| Mohammad Hamza | UI/UX Designer         | hamza.mohammad@email.com | +92 303 4567890 | View    | Day 3 ▼ |
| Zara Noor      | QA Engineer            | zara.noor@email.com      | +92 304 5678901 | View    | Day 2 ▼ |
| Ali Raza       | Database Administrator | ali.raza@email.com       | +92 305 6789012 | View    | Day 1 ▼ |
| Sana Javed     | Product Manager        | sana.javed@email.com     | +92 306 7890123 | View    | Day 3 ▼ |

Figure 8 shows Assign Different Days to Candidates submodule

#### Validators List:

- Select Day Dropdown: A valid interview day must be selected
- Day Assignment per Candidate: Must select from available scheduled days (Day 1, Day 2, etc.)
- Search Field: Accepts text input; trims whitespace
- View Button: Only enabled when a candidate row is selected

### 16.1.1 Manage Candidates

**Candidates Management**

Export

+ Insert New Candidates   Positions   Advertisement   Days   Search...

| ID | Candidate Name | Position               | Email                    | Phone           | Applied Date | Actions   |
|----|----------------|------------------------|--------------------------|-----------------|--------------|-----------|
| 1  | Ayesha Khan    | Software Engineer      | ayesha.khan@email.com    | +92 300 1234567 | 10 May 2025  | Edit View |
| 2  | Usman Ali      | System Analyst         | usman.ali@email.com      | +92 301 2345678 | 11 May 2025  | Edit View |
| 3  | Hina Fatima    | DevOps Engineer        | hina.fatima@email.com    | +92 302 3456789 | 12 May 2025  | Edit View |
| 4  | Mohammad Hamza | UI/UX Designer         | hamza.mohammad@email.com | +92 303 4567890 | 13 May 2025  | Edit View |
| 5  | Zara Noor      | QA Engineer            | zara.noor@email.com      | +92 304 5678901 | 14 May 2025  | Edit View |
| 6  | Ali Raza       | Database Administrator | ali.raza@email.com       | +92 305 6789012 | 15 May 2025  | Edit View |
| 7  | Sana Javed     | Product Manager        | sana.javed@email.com     | +92 306 7890123 | 16 May 2025  | Edit View |
| 8  | Danish Ahmed   | Business Analyst       | danish.ahmed@email.com   | +92 307 8901234 | 17 May 2025  | Edit View |
| 9  | Maryam Hashmi  | Mobile App Developer   | maryam.hashmi@email.com  | +92 308 9012345 | 18 May 2025  | Edit View |
| 10 | Farhan Saeed   | Network Engineer       | farhan.saeed@email.com   | +92 309 0123456 | 19 May 2025  | Edit View |

Figure 9 shows Candidates Management module

#### Validators List:

- Candidate Name: Must not be empty; no special characters
- CNIC: Must be exactly 13 digits; numeric only
- Position Filter Dropdown: Must select a valid position
- Advertisement Filter Dropdown: Must select a valid advertisement
- Days Filter Dropdown: Must select a valid interview day
- Insert New Candidates Button: All mandatory fields must be filled before saving
- Search Field: Accepts text input; trims whitespace

### 17.1.1 Mark Attendance

**Candidate Attendance**  
Available Advertisements

Export Options ▾

Search...

| Advertisement Title      | Advertisement ID | Department            | Posted Date | Actions |         |                 |
|--------------------------|------------------|-----------------------|-------------|---------|---------|-----------------|
| Senior Software Engineer | ADV-2025-001     | IT Department         | 01 May 2025 | View    | Day 1 ▾ | Mark Attendance |
| System Analyst           | ADV-2025-002     | IT Department         | 02 May 2025 | View    | Day 2 ▾ | Mark Attendance |
| HR Executive             | ADV-2025-003     | HR Department         | 03 May 2025 | View    | Day 1 ▾ | Mark Attendance |
| DevOps Engineer          | ADV-2025-004     | IT Department         | 04 May 2025 | View    | Day 3 ▾ | Mark Attendance |
| UI/UX Designer           | ADV-2025-005     | Design Department     | 05 May 2025 | View    | Day 2 ▾ | Mark Attendance |
| QA Engineer              | ADV-2025-006     | IT Department         | 06 May 2025 | View    | Day 1 ▾ | Mark Attendance |
| Data Analyst             | ADV-2025-007     | IT Department         | 07 May 2025 | View    | Day 2 ▾ | Mark Attendance |
| Product Manager          | ADV-2025-008     | Product Department    | 08 May 2025 | View    | Day 3 ▾ | Mark Attendance |
| Marketing Executive      | ADV-2025-009     | Marketing Department  | 09 May 2025 | View    | Day 1 ▾ | Mark Attendance |
| Business Analyst         | ADV-2025-010     | Operations Department | 10 May 2025 | View    | Day 2 ▾ | Mark Attendance |

Figure 10 shows Candidate Attendance module

#### Validators List:

- **Advertisement Selection:** A valid advertisement must exist
- **Day Dropdown:** A valid scheduled day must be selected before marking attendance
- **Search Field:** Accepts text input; trims whitespace
- **Mark Attendance Button:** Only active when advertisement and day are selected



## 17.1.2 Mark Attendance

**Candidate Attendance**  
Mark Candidate Attendance

Export

Search...

| ID | Candidate Name | Position          | Date        | Time     | Status  | Remarks         | Actions   |
|----|----------------|-------------------|-------------|----------|---------|-----------------|-----------|
| 1  | Ayesha Khan    | Software Engineer | 12 May 2025 | 09:15 AM | Present | On time         | Present ▾ |
| 2  | Ali Raza       | Data Analyst      | 12 May 2025 | 09:30 AM | Absent  | Sick leave      | Absent ▾  |
| 3  | Usman Tariq    | QA Engineer       | 12 May 2025 | 09:05 AM | Present | On time         | Present ▾ |
| 4  | Sara Ahmed     | UI/UX Designer    | 12 May 2025 | 09:20 AM | Present | On time         | Present ▾ |
| 5  | Hassan Ali     | DevOps Engineer   | 12 May 2025 | 09:45 AM | Absent  | Personal reason | Absent ▾  |
| 6  | Maham Noor     | System Analyst    | 12 May 2025 | 09:10 AM | Present | On time         | Present ▾ |
| 7  | Bilal Hussain  | Business Analyst  | 12 May 2025 | 09:35 AM | Absent  | Traffic delay   | Absent ▾  |
| 8  | Zainab Fatima  | HR Executive      | 12 May 2025 | 09:00 AM | Present | On time         | Present ▾ |
| 9  | Danish Mehmood | Network Engineer  | 12 May 2025 | 09:55 AM | Absent  | Out of city     | Absent ▾  |
| 10 | Iqra Saleem    | Content Writer    | 12 May 2025 | 09:25 AM | Present | On time         | Present ▾ |

Figure 11 shows Mark Attendance submodule

### Validators List:

- Attendance Status per Candidate: Must select “Present” or “Absent” from dropdown
- Search Field: Accepts text input; trims whitespace
- Save Action: Not allowed until all candidates have an attendance status selected

### 18.1.1 Candidate Evaluation

**Candidate Evaluation**

Available Advertisements

Export Options

Search...

| ID | Advertisement Title      | Position          | Posted On   | Last Date   | Actions |         |                      |
|----|--------------------------|-------------------|-------------|-------------|---------|---------|----------------------|
|    |                          |                   |             |             | View    | View By | Action               |
| 1  | Software Engineer Hiring | Software Engineer | 01 May 2025 | 20 May 2025 | View    | Day ▾   | Interview Candidates |
| 2  | Data Analyst Opening     | Data Analyst      | 02 May 2025 | 21 May 2025 | View    | Day ▾   | Interview Candidates |
| 3  | QA Engineer Required     | QA Engineer       | 03 May 2025 | 22 May 2025 | View    | Day ▾   | Interview Candidates |
| 4  | UI/UX Designer Vacancy   | UI/UX Designer    | 04 May 2025 | 23 May 2025 | View    | Day ▾   | Interview Candidates |
| 5  | DevOps Engineer Role     | DevOps Engineer   | 05 May 2025 | 24 May 2025 | View    | Day ▾   | Interview Candidates |
| 6  | System Analyst Position  | System Analyst    | 06 May 2025 | 25 May 2025 | View    | Day ▾   | Interview Candidates |
| 7  | Business Analyst Hiring  | Business Analyst  | 07 May 2025 | 26 May 2025 | View    | Day ▾   | Interview Candidates |
| 8  | HR Executive Needed      | HR Executive      | 08 May 2025 | 27 May 2025 | View    | Day ▾   | Interview Candidates |
| 9  | Network Engineer Role    | Network Engineer  | 09 May 2025 | 28 May 2025 | View    | Day ▾   | Interview Candidates |
| 10 | Content Writer Position  | Content Writer    | 10 May 2025 | 29 May 2025 | View    | Day ▾   | Interview Candidates |

Figure 12 shows Candidate Evaluation module

#### Validators List:

- Advertisement Selection: A valid advertisement must be listed
- Day Dropdown per Row: A valid interview day must be selected
- Search Field: Accepts text input; trims whitespace
- Interview Candidates Button: Only enabled when advertisement and day are selected



## 18.1.2 Interview Candidate

The screenshot displays the 'Interview Candidates' submodule. On the left is a dark sidebar menu with options: Dashboard, Positions, Advertisement, Intelligent Screening, Interviews (highlighted), Candidates, Attendance, Evaluation, and Experience Calculator. The main content area has a title 'Interview Candidates' and a subtitle 'Mark Candidate Attendance'. There is an 'Export' button and a search bar. Below is a table with 10 rows of candidate data.

| ID | Candidate Name | Interview Date & Time | Interviewer   | Remarks         | Comments                      |
|----|----------------|-----------------------|---------------|-----------------|-------------------------------|
| 1  | Ayesha Khan    | 12 May 2025, 09:00 AM | Sarah Ali     | Recommended     | Good communication skills     |
| 2  | Ali Raza       | 12 May 2025, 10:00 AM | Usman Tariq   | Not Recommended | Needs technical improvement   |
| 3  | Usman Tariq    | 12 May 2025, 11:00 AM | Sarah Ali     | Recommended     | Strong problem solving        |
| 4  | Sara Ahmed     | 12 May 2025, 12:00 PM | Bilal Hussain | Null            | Did not appear                |
| 5  | Hassan Ali     | 13 May 2025, 09:00 AM | Usman Tariq   | Recommended     | Good analytical skills        |
| 6  | Maham Noor     | 13 May 2025, 10:00 AM | Sarah Ali     | Not Recommended | Lacks experience              |
| 7  | Bilal Hussain  | 13 May 2025, 11:00 AM | Usman Tariq   | Recommended     | Excellent technical knowledge |
| 8  | Zainab Fatima  | 13 May 2025, 12:00 PM | Sarah Ali     | Null            | Candidate unavailable         |
| 9  | Danish Mehmood | 14 May 2025, 09:00 AM | Bilal Hussain | Recommended     | Good attitude                 |
| 10 | Iqra Saleem    | 14 May 2025, 10:00 AM | Usman Tariq   | Not Recommended | Needs more confidence         |

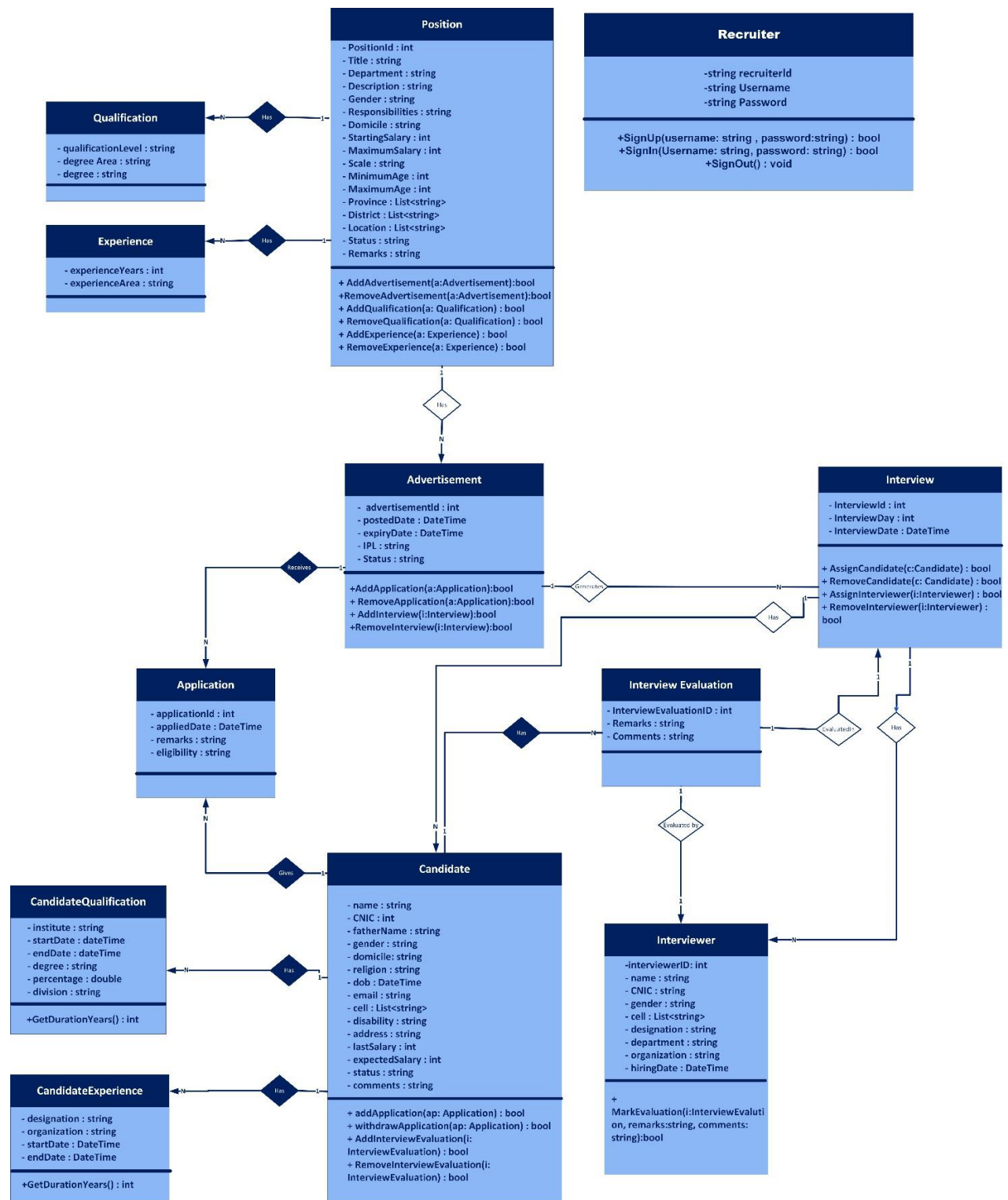
Figure 13 shows Interview Candidate submodule

### Validators List:

- Remarks Dropdown per Candidate: Must select from allowed values (Recommended, Not Recommended, Null)
- Comments Field: Optional; free text; max 500 characters recommended
- Search Field: Accepts text input
- Save Action: Remarks must be set for all visible candidates before saving

## 7. UML Diagrams

### 7.1 Class Diagram



## 8. Weakness & Future Improvements

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The system shortlist candidates only based on the details provided by the candidate considering him honest, It does not support uploading the documentations of candidates to verify the authenticity of the claim of the candidate. In future we will add this feature and integrate artificial intelligence to evaluate the authenticity of candidate's claim and to verify the documents provided but the candidates.